

DATA QUALITY ASSURANCE OF CO₂ STORAGE DATA 2026

Data quality assurance - 2026

Imperial College London Ltd.

Report no.: 10637467, Rev. 0 DRAFT

Document no.: 3093337

Date: 2026-09-01





Project name:	Data Quality Assurance of CO2 Storage Data 2026	DNV Services UK Limited Energy
Report title:	Data quality assurance - 2026	Systems
Customer:	Imperial College London Ltd., Exhibition Road Sherfield	Data Consulting
	Building SW7 2AZ London	5th Floor Vivo Building Vivo Building
	United Kingdom	30 Stamford Street SE1 9LQ
Customer contact:	Xiaowei Gao	London
Date of issue:	2026-09-01	United Kingdom
Project no.:	10637467	Tel:
Organisation unit:	Data Consulting	GB 440 60 13 95
Report no.:	10637467, Rev. 0 DRAFT	
Document no.:	3093337	
Applicable contract(s) governing the provision of this Report:		

Objective:

Prepared by:

Verified by:

Approved by:

Marco Virgili
Senior Consultant

Dave Simpson
Team Lead, Data Consulting

DNV information classification and internal sharing:

Information classification in DNV:

- ☐ Open
☐ DNV Restricted
☐ DNV Confidential
☐ DNV Secret

Internal sharing permitted in DNV:

Yes (default)

☐ No ☐ Yes

Additional authorized approvers:

Keywords

Rev. no.	Date	Reason for issue	Prepared by	Verified by	Approved by
0		First issue			

Copyright © DNV 2026. All rights reserved. Unless otherwise agreed in writing: (i) This publication or parts thereof may not be copied, reproduced or transmitted in any form, or by any means, whether digitally or otherwise; (ii) The content of this publication shall be kept confidential by the customer; (iii) No third party may rely on its contents; and (iv) DNV undertakes no duty of care toward any third party. Reference to part of this publication which may lead to misinterpretation is prohibited.

DISCLAIMER

Independence, impartiality, and advisory limitations

This document contains content provided by DNV. Please note the following:

Ethical safeguards

To maintain integrity and impartiality essential to its third-party roles, DNV performs initial conflict-of-interest assessments before engaging in advisory services.

Priority of roles

This report is generated by DNV in its advisory capacity, subsequent to conflict-of-interest assessments. It is separate from DNV's responsibilities as a third-party assurance provider. Where overlap exists, assurance activities conducted by DNV will be independent and take precedence over the advisory services rendered.

Future assurance limitation

The content in this document will not obligate or influence DNV's independent and impartial judgment in any future third party assurance activities with DNV.

Compliance review

DNV's compliance with ethical and industry standards in the separation of DNV's roles is subject to periodic external reviews.

TABLE OF CONTENTS

1	EXECUTIVE SUMMARY	1
2	INTRODUCTION	2
2.1	Objectives	2
2.2	Past cycles	2
3	INDUSTRY CONTEXT	3
3.1	Reporting standards	3
4	ASSESSMENT METHODOLOGY	4
4.1	Syntactic quality	4
4.2	Semantic quality	5
4.3	Pragmatic quality	5
4.4	Data profiling	5
4.5	Areas of evaluation	5
5	ASSURANCE BASIS.....	7
5.1	Overview	7
5.2	Core requirements	7
5.3	Data quality principles	7
5.4	Assurance plan	8
6	EVALUATION	9
6.1	Data profiling	9
6.2	Updates since 2025 review	10
6.3	Project data workflow	11
6.4	Data Validation	12
6.5	Data Structure	13
6.6	Source data	13
7	FINDINGS AND RECOMMENDATIONS.....	14
7.1	Findings	14
7.2	Archiving data	14
7.3	Organisational Maturity	15
8	REFERENCES	17
	APPENDIX A – DATA VALIDATION RULES	18
	APPENDIX B – DATA QUALITY CHECKS	19
	APPENDIX C – ORGANISATION MATURITY	20

1 EXECUTIVE SUMMARY

The London Register of Subsurface CO₂ Storage (“the register”) is a pioneering initiative that consolidates project-level global CO₂ storage data from operational carbon capture and storage (CCS) projects. This register aims to address the critical need for accessible, trustworthy, and comprehensive data to support scientific research, policy making, investment decisions, and public transparency in the rapidly evolving CCS sector.

This assurance review, conducted by DNV, evaluates the register’s data quality and management practices with reference to internationally recognised standards, including DNV-RP-0497 and ISO 8000 by building upon the foundations set by the previous review. Prioritised data quality principles have been confirmed—completeness, understandability, traceability, and accuracy—as the foundation for assessing the register’s value and reliability. DNV has reviewed the revised register against the prioritised principles, providing recommendations for future development and enhancement.

The register demonstrates strong coverage and transparency, being available for peer review and benchmarking. Several improvements have been made to the way the data is managed and validated, which represent a significant improvement in terms of organisational maturity.

Further opportunities for enhancement of the register and associated documents have been noted. For example, the addition of automated validation checks on the database itself, which would mitigate the risk of inaccurate entries. The explicit connection to a source document archive would enhance the transparency of the process, as publishers can edit the sources at any time, impacting the ability for users to verify the accuracy of the register. Other areas of improvement include the adoption of some additional data engineering best practices, more extensive logging, and additional clarifications in the methodology document.

Such observations do not detract from the progress made in adopting the recommendations in the previous assessment cycle, but are focussed on further enhancing the register’s quality.

2 INTRODUCTION

The London Register of Subsurface CO₂ Storage (hereafter “the register”) compiles project-level global CO₂ storage data from operational CCS projects on an annual basis. Storage data is sought for many purposes by the scientific community, policy makers, technology providers, and the public. Such purposes include tracking industry progress, project performance and global trends; which in turn inform insight, investment and policy decisions. Access to trustworthy data is valuable to support these activities.

Most operating storage projects report their storage rates publicly; however, these sources vary widely in format, publication type, and reporting methods. This leads to challenges in accessing and analysing global data; hence the register seeks to alleviate access and usability challenges by presenting a consolidated source of global data. It is worth noting at the outset that the aim of the register to provide publicly available, high value data are commendable.

The previous review /1/ established prioritised data quality principles which represent the objectives and value of this dataset. The register has subsequently been reviewed against these principles. This has been carried out based on DNV-RP-0497 “Assurance of data quality management” /2/.

2.1 Objectives

Data assurance provides confidence that data meets a specific need, and that its requirements have been or will be achieved. DNV has carried out an assurance review of the register to evaluate the extent to which it fulfils its requirements and make recommendations for enhancement.

The first version of the register was published in November 2025 /3/, and the publication of the next version is planned for November 2026. In this report we review the data gathering process, format and contents of the register against the data quality principles established. Reference is made to internationally recognised data quality standards, including DNV-RP-0497 and ISO 8000 /4/.

2.2 Past cycles

An initial assessment was carried out prior to the initial register publication in November 2025. Since the previous cycle there have been significant updates made with the aim of improving the data quality, and DNV has been provided with evidence of these changes by the register team. This report reviews these changes, and aims to identify further areas for improvement, with the goal of helping the register achieve an even higher level of maturity over time.

3 INDUSTRY CONTEXT

DNV's ETO "CCS to 2050" /5/ report states:

"a consistent approach to reporting operational performance and transparency regarding the data could offer significant benefits to the CCS industry. Such data could enable more accurate quantification of CO₂ avoided and provide the basis for bench-marking and performance improvements"

In CCS, as for all components of the energy transition, achieving net zero requires clear strategy, policy alignment, investment, and global coordination: all of which are built upon a bedrock of quality data. The efforts of the London Register of Subsurface CO₂ storage in seeking to provide a consolidated view of global storage data are an important contribution to this critical foundation.

With more CCS projects coming into operation there is a need for transparent reporting of project performance, particularly to understand the impact of CCS as a climate change mitigation tool. As well as this, trust in reported volumes of CO₂ storage is critical when considering commercial implications associated with compliance and voluntary carbon markets, subsidy support schemes and tax credits.

Global CO₂ storage data can be used to track and compare industry progress across regions, influencing policy and technology development, and informing models and predictions in the effort to tackle climate change. Without high quality data these aims are inhibited, and risks are introduced, including:

- Reputational damage, to operators, reporting bodies, and the industry as a whole.
- Misinformed and suboptimal decision making.
- Loss of confidence in CO₂ storage technology, operators and industry.
- An unrealistic picture of the role of CO₂ storage, an inability to forecast and track the contribution to climate change.

For data to have impact it must be trustworthy, and it must be trusted. This is represented in the data's quality characteristics, reflecting the data sources, gathering and transformation processes, presentation format, and means of access.

3.1 Reporting standards

The register consolidates data from a variety of sources. Some of the data sources reference reporting standards, which give context to the data reported. A summary of the various standards is provided in the methodology document /6/.

The reporting standards provide useful context for understanding the origin, scope and intended use of the reported data. However, the standards vary considerably in their objectives, definitions, reporting requirements and assurance processes. Some are focused specifically on CO₂ storage activities, while others form part of broader corporate sustainability or greenhouse gas reporting frameworks.

As a result, the presence of a reporting standard should not be interpreted as a consistent indicator of data quality or reliability across all records in the register. Differences between standards may also influence how certain metrics, including CO₂ storage rates, are defined and reported. Therefore, quantifying the uncertainty linked to each standard has a subjective component that relies on expert judgement. This is acknowledged in the methodology document.

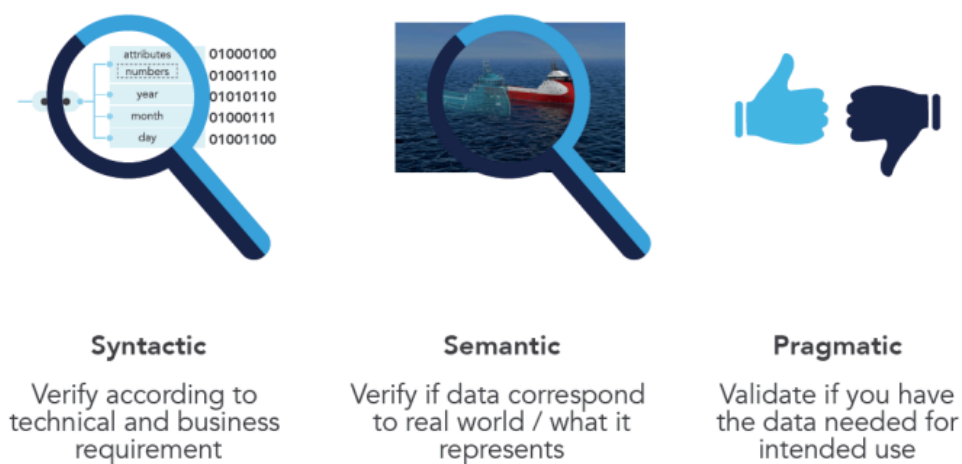
4 ASSESSMENT METHODOLOGY

DNV's recommended practice DNV-RP-0497 "Assurance of data quality management", provides a framework for the assessment of data quality and data quality management. This is firmly rooted in ISO 8000, providing a practical application of data quality principles detailed in this ISO series.

ISO 8000 defines fundamental concepts of data quality and their relationship to quality management systems. The standard defines three categories for data quality:

- **Syntactic:** The degree to which data conforms to their specified syntax.
- **Semantic:** The degree to which data corresponds with that which it represents.
- **Pragmatic:** The degree to which data is appropriate and useful for a particular purpose.

Figure 4-1: ISO 8000-8 data quality assessment model



4.1 Syntactic quality

Syntactic quality measures the extent to which a dataset conforms to its declared structure — including schema, metadata, and pattern requirements. As defined by DNV-RP-0497 and ISO 8000, this involves checking syntactic and integrity constraints against a defined schema and rule base, supported by data profiling to identify illegal values, outliers, or structural anomalies before semantic or pragmatic assessment. The approach is bottom-up and often tool-assisted, and may include steps to:

1. Establish or confirm the reference schema and conceptual model.
2. Profile the data to reconstruct missing metadata and identify inconsistencies.
3. Express syntactic requirements as executable rules (e.g. for completeness, uniqueness, type/format, domain lists, allowed ranges, precision/scale, and temporal bounds); and
4. Perform automated checks to derive metrics (e.g., pass/fail counts, defect rates) and produce flagged-record lists for remediation.

To carry out these quality checks, syntactic requirements should be clearly defined and measurable. The register currently does not include syntactic definitions. It is recommended that such a specification be developed, to aid understanding and enable definition of quality checks and rules, see recommendations in Section 7.

4.2 Semantic quality

Semantic quality assesses how accurately the data reflects the real-world concepts it represents. While syntactic checks confirm structure, semantic checks confirm meaning. This ensures that values are consistent, logical, and aligned with reference sources.

The assessment follows a model-driven approach:

1. Define the semantic model and relevant reference sources.
2. Profile the data to identify inconsistencies or implausible values with respect to physical reality.
3. Implement executable rules to test accuracy, consistency, and coherence; and
4. Generate metrics and exception lists for review and remediation.

The above-specified activities implement the semantic data quality considerations from the DNV-RP-0497 framework, ensuring that both meaning and relationships are consistent, traceable, and verifiable throughout the dataset.

4.3 Pragmatic quality

The pragmatic quality reflects the degree to which the data is considered fit for its intended use — being complete, timely, and reliable enough to support decisions or analyses. While *syntactic* checks confirm structure and *semantic* checks confirm meaning, pragmatic checks evaluate usefulness within the application context.

The assessment takes into account both the operational and decision-making context of the dataset. It examines whether the available information is sufficient to meet the stated purpose, whether appropriate confidence levels are assigned, and whether any known limitations affecting its use are documented. Typical considerations include data coverage, temporal consistency, stability, and reproducibility of derived values.

Pragmatic checks confirm that the dataset can be used with an agreed level of confidence for its objectives, highlighting where supplementary data or clarification may be required.

4.4 Data profiling

Data profiling is a valuable and commonly used method for exploring the characteristics of datasets, particularly where metadata is limited or absent. It enables the empirical investigation of statistical patterns and the identification of potential quality issues such as illegal values, outliers, and other anomalies. Through this process, metadata can be re-engineered, providing deeper insights into the dataset's composition and integrity.

Data profiling is often used in the early stages of data activities, such as migrations, integrations, and analytics to anticipate whether actions might be necessary to address the quality issues. For the register data, the results of the profiling can form the basis of syntactic quality requirements. Profiling has also informed generation of indicative data quality rules, see Appendix A.

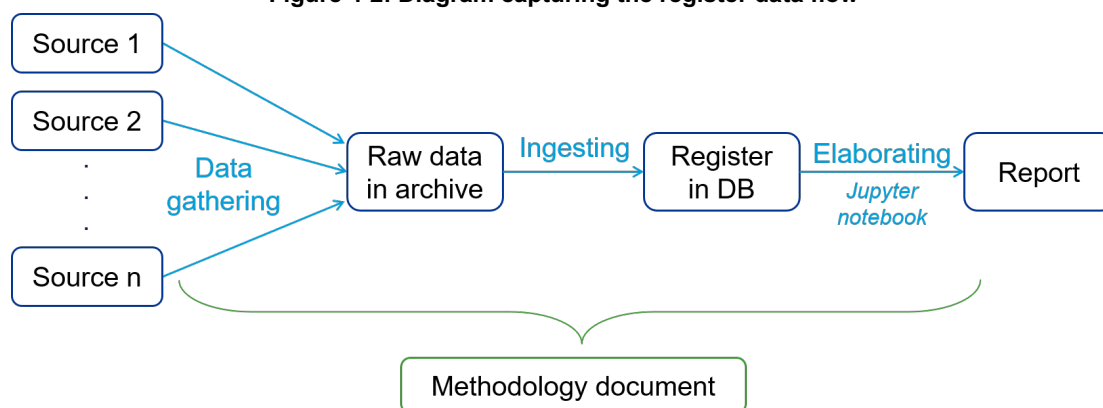
Section 6 presents the outputs of the data profiling activities conducted. These included structural checks such as data types, maximum length and precision, as well as key integrity indicators like primary key flags.

4.5 Areas of evaluation

The data lifecycle, as captured in Figure 4-2, represents all the steps the data goes through before being saved into the register and then used to produce the report: first, the data sources are identified and data is extracted from them, then data is ingested into the database, and finally aggregated to produce the figures listed in the report. DNV's assessment covered these steps methodically. The assessment included the following phases:

1. A sample of projects' data has been reviewed to evaluate the data lineage and clarity of presentation in the register.
2. A review of the data engineering principles reflected in the register database.
3. A comparison between the data in the Excel sheets extracted from the database and the data in the database to evaluate the extraction process.
4. Review of the use of data visualisation in the report, with traceability comparison to the raw data in the register.
5. A review of the register methodology by DNV's subject matter experts to identify any critical aspects specific to the field of CO₂ storage accounting.
6. A review of the methodology document /6/ released by the register team for consistency and completeness.
7. An evaluation of the data quality characteristics of the register against the rules described in the Assurance Basis.

Figure 4-2: Diagram capturing the register data flow



Furthermore, the register team asked DNV for:

- a recommendation on how to deal with the uncertainty introduced by the variety of reporting standards;
- a recommended strategy for maintaining a public archive of past data sources directly linked to the database.

The former aims to help contextualise any takeaways by accompanying values with confidence intervals, while the latter is a way to address the issue of source documents changing or becoming unavailable when publishers update them.

The register team identified these as opportunities for improvement and asked DNV for recommendations on how to tackle them. These have been addressed in section 6 and section 7.

A list of the individual checks carried out by DNV, categorised by data quality principle, is reported in Appendix B.

5 ASSURANCE BASIS

5.1 Overview

An assurance basis was established during the previous review cycle to define the intended purpose, key requirements and quality objectives of the register. This provides the framework against which assurance activities are performed and findings assessed.

An assurance basis consists of an agreed framework within which an assurance review can be carried out. This is critical to the establishment of the “claims” against which the solution can be reviewed, with reference to evidence items. The assurance approach is outlined in DNV-RP-0497, which provides a systematic approach and framework to assuring data quality.

The assurance basis remains applicable for the current review and was confirmed with the register team as continuing to reflect the intended use of the register. However, its scope was expanded to include the documents and assets that were not part of the previous iteration of the register.

Full details of the assurance basis, including the workshop process and supporting standards references, are provided in the 2025 data quality report /1/.

5.2 Core requirements

The foundational motivation for establishing the database was the difficulty in gathering data from disparate sources for any holistic view of global CO₂ storage. The originating requirement of the register is therefore that it is as comprehensive as possible: covering all operational CO₂ storage projects globally; and that it is sufficiently transparent to promote trust and therefore use. The following general requirements apply:

1. Completeness: global coverage, enriched with layered project data and accompanying metadata.
2. Usability and understandability: users are able to retrieve and understand the data sufficiently for their use; this includes a structure favouring flexibility and future scalability.
3. Transparency and traceability: full data lineage is readily available promoting trust and enabling use-case decision making (including limitations).
4. Data integrity and accuracy: ‘trueness’ enabling robust and quality decision making (includes considerations for uncertainty quantification).

5.3 Data quality principles

The core requirements presented above can be represented in the following principles:

1. The data is sufficiently comprehensive, covering all major operational subsurface CO₂ storage projects globally.
2. The data is sufficiently well documented and structured so that a target user can utilise it easily, and without additional support.
3. The data sources and processing methodology are sufficiently well documented to inspire trust in the broad community and enable understanding of limitations for individual use cases.
4. The data reflects the truest available representation of injection rates for each project.

5.4 Assurance plan

The register was initially published in November 2025. Major changes have been made to its management, structure, and contents over the past year; most of the recommendations provided in the previous cycle have been actioned. The impact of these changes is part of this review.

Recommendations made in the previous review focussed on the foundations necessary for a robust and transparent data aggregation process. In addition to assessing the impact of these changes, this review focusses on identifying targeted improvements and enhancing best practice for the future. The four data quality principles were used to develop a list of checks, which are reported in Appendix A, with the associated requirement and type (qualitative or quantitative).

The rules are applied to all the datasets and fields where they are applicable, and the results of the evaluation are reported in Section 6.

6 EVALUATION

6.1 Data profiling

This review was based on the following material, soon-to-be published:

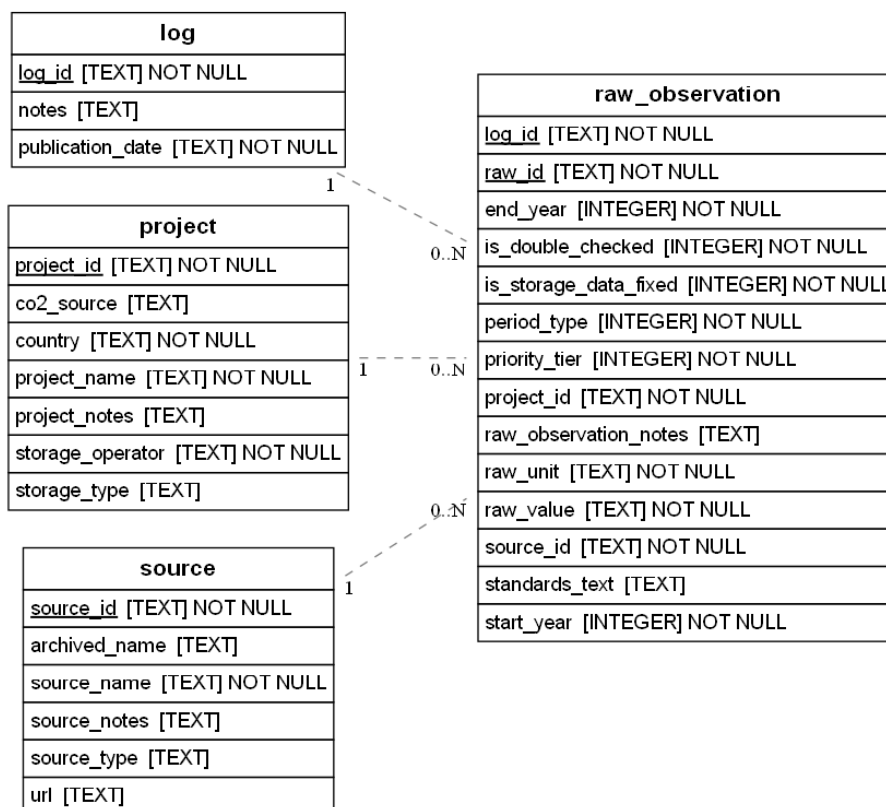
- A SQLite extraction /7/ of the original register database, in .db format.
- The register annual report /8/.
- The register methodology document /6/.
- An Excel spreadsheet /9/ containing processed register data.

The register team also supplied the following additional information to DNV:

- A response document /10/ describing the actions taken to address the recommendations provided by DNV in the previous review cycle.
- Four database extracts in .csv format /11//12//13//14/, each capturing a database table (log, project, raw_observation, and source).
- The Jupyter Notebook script /15/ used to process the register data and generate the report graphs, based on the csv extracts.

For the purposes of assurance, the core register artefact is the relational database. An entity relationship diagram (ERD) for the provided SQLite database is shown in Figure 6-1.

Figure 6-1: ERD for SQLite database



6.2 Updates since 2025 review

6.2.1 Relational database

The recommendation to store the data in a relational database has been implemented. The data is now held within a structured database comprising linked tables, with the published spreadsheet report generated from a csv extract of such tables, using a Python script. In addition, all supporting information is now stored in dedicated columns rather than being additionally represented through formatting and annotations, as was the case in the previous version.

6.2.2 Separation of raw processed data

The database has been designed to preserve source values in their original form, together with the units in which they were reported. The transformations and processing required to generate the final report are performed separately using the provided Jupyter Notebook script. This approach maintains a clear distinction between raw source data and processed outputs.

6.2.3 Quality control

Quality control measures have been implemented as recommended. The database includes flags to indicate whether an entry has undergone secondary review, and this check has been completed for most data inputs. The methodology document mentions that the quality assurance is carried out by a dedicated role, though the role is not formally described. In addition, the register team circulates draft outputs to the consortium for further review and validation before publication.

6.2.4 Data processing log

The database includes a log table that records each version together with a description of the changes made. However, detailed change tracking between versions is not currently available: only the values associated with the latest version are retained, so it is not possible to compare historical versions or identify the specific data changes made between releases.

6.2.5 Methodology documentation

The methodology document published alongside the report provides a comprehensive description of the approach used. The accompanying Jupyter Notebook, which performs the data processing, is well documented and contains sufficient annotation to allow users to understand, review, and reproduce the published report. The script was successfully tested and can recreate the released report. However, a few inconsistencies were spotted between the methodology document and some of the register artefacts, for example a definitive list of the published register artefacts, a complete description of the roles in the project, and some of the reported figures. While data dictionaries are included for key fields in the documentation, a full data dictionary should be developed, maintained, and published. This enhances useability and reduces the risk of misinterpretation of the database, as well as aiding future development and maintenance of the database by the register team.

6.2.6 Presentation standards

The database and report accurately reflect the presentation rules described in the methodology (i.e., consistent units and number of digits) and adhere to the prioritisation scheme defined in the methodology document. This is enabled by the validation rules discussed in Section 6.4. The data gaps in the most recent year of data are also shown in the diagrams to highlight areas where data currently hasn't been obtained, though the methodology adopted to do so, as well as the underlying assumptions, are not documented in the report.

6.2.7 Data quality rules

The Jupyter Notebook includes several automated validation checks implemented through raise error functions, which assess data against some of the rules set out in the previous report. These checks provide a useful mechanism for identifying data quality issues during processing. These validation checks consist of: valid units of measure, duplicated entries, positive value increments, missing projects at aggregation, and negative aggregate residuals. The validation

process would benefit from a more extensive ruleset, and from being applied directly to the database, not only during data processing.

6.2.8 Reporting standards

The changes to the reporting standards have been partially implemented: the report now includes an appendix that catalogues all reporting standards and frameworks used in the calculation of CO₂ storage values, together with their classifications, descriptions, and links to the relevant source material. Furthermore, the accompanying summary table provides additional context by showing the countries in which each framework is referenced, along with the number of associated projects and observations. The timeline figure is particularly effective in illustrating the evolution and version history of each framework. The documentation acknowledges the lack of a straightforward relationship between a cited reporting standard and the confidence that can be placed in a specific project-year value. Furthermore, the standards do not always provide a definition of storage and to what extent each source respects the reference standard. Therefore it is not currently possible to infer the meaning of these references in interpretation of the register.

6.2.9 Source data gathering

The findings identified during the sample data check in the previous review have been addressed through enhancements to both the database design and the structure of the published report. The database now preserves source values in their original form, together with the units in which they were reported. Issues relating to potential double counting across reporting periods and data sources have been addressed through an improved methodology for handling cumulative values, supported by a database structure that includes source prioritisation fields to distinguish between alternative data sources. In terms of completeness, whilst it cannot be guaranteed that all projects are included, there is now three-layered process in place to include all known projects. Additional improvements include clearer documentation of the channels through which survey responses were received and the inclusion of notes to explain observations that differ from the original source material.

6.2.10 Documentation

The published methodology, database containing raw values, and the data processing script have aided in the traceability and understandability register. The structure of the processing script and database as well as the review flags enhance the quality control process and therefore accuracy of the register outputs. All the documents are structured logically but do not employ dynamic cross-referencing, which caused some internal references to become outdated.

6.3 Project data workflow

The data workflow, as described in the methodology document, consists of four phases:

1. Scoping and source identification.
2. Data collection and standardisation.
3. Review, validation, and assurance.
4. Publication and archiving.

This section is dedicated to reviewing each of these to recognise good practice and identify opportunities for further improvement.

6.3.1 Scoping and source identification

Identifying all existing CO₂ storage projects worldwide is a crucial part of the register's goal. This is achieved through three layers of checks:

1. Capture the latest list of CO₂ storage projects published by the Global CCS Institute.

2. Ask consulting partners for input on projects that may be missing from the list.
3. Hold a series of webinars to ask experts in the field for further input on missing projects.

The register team is aware that no process is guaranteed to identify all the existing projects because not all countries may be willing to share what they may see as strategic information. However, the three-layered process is a robust approach.

6.3.2 Data collection and standardisation

The data is manually extracted from each source after downloading the relevant files, and ingested into the database, converting information to fit into the database schema. This includes the creation of unique identifiers for each row of each table.

If a value is available from multiple sources, these are assigned a priority score in order to capture all available information without creation of duplicates. This is a useful way of logging all available information without overestimating aggregated values.

6.3.3 Review, processing, and assurance

Most data entries are quality assured through a four-eyes review. Two fields in the database are dedicated to flag entries that have undergone this process (“is_double_checked” and “is_storage_data_fixed”), and the process is captured in the methodology document. Section 6.6 contains a review of the database against the original sources to assess the robustness of the QA process.

Once in the database, data is validated and processed using the Jupyter Notebook. While this process accurately follows the documented methodology, the code could benefit from some restructuring to enhance readability. For example, not all functions have docstrings, some loops are condensed into a single line, spacing is inconsistent, and there are instances of hard-coded values. Furthermore, the validation checks do not capture all the potential validation rules, as listed in the last report and in Appendix A, and these are only applied at this processing stage. Additional validation rules such as year-on-year change in value, the comparison with an expected list of values, and the flagging of any gaps in coverage may direct manual review to entries that may be inaccurate. Moving these checks out of the processing script and running them on the database directly would allow continuous evaluation of the core data quality and allowing tracking over the register’s lifecycle.

6.3.4 Publication and archiving

The processed data is analysed and discussed in the report, with the aid of a set of graphs visualising key insights drawn from the analysis. The graphs complement the written text as a means of conveying information beyond the values, however they may be more impactful with a targeted design.

Outdated sources are replaced by new ones and archived with consistent names, which is good practice for traceability, but they are not linked to the database with unique identifiers. The archiving strategy recommended by DNV is described in Section 7.2 of this report. The archive is not publicly accessible at the time of writing, though this is being considered by the register team and reflected in the methodology document.

6.4 Data Validation

Data validation is currently carried out during data processing, using the Jupyter Notebook to raise flags when validation rules are broken. The presence of such checks is a valuable addition to the process, but they could be further expanded in scope. To offer guidance in this area, DNV developed a list of data validation rules based on the ones provided in the previous report, adapted to the new data format and expanded with new checks. The resulting list of rules is reported in Appendix A, categorised by data quality requirement.

As part of this review, DNV have screened the database against these validation rules, with the following observations:

- Sudden increases in reported CO₂ storage or gaps in years covered, usually linked to a change in reporting conventions.
- Missing reporting standards or source URLs.
- Projects being active for a short period of time (less than two years).

While none of the observations point to obvious errors in the data ingestion, monitoring these aspects can help spotting any changes early in the review process, as well as giving initial insight into notable results.

6.5 Data Structure

The review of the database structure has highlighted the following observations:

- **Technology stack:** the database is provided as a .db file, which contains a SQLite database, however the methodology document refers to Microsoft SQL Server. The methodology document refers to the .db file as a “build script” and does not specify how this is extracted from the SQL Server.
- **Documentation:** some versions of the register are recorded in the database log table but without accompanying data. There is information in the methodology document describing the database, including the data dictionaries and the schema diagram, which have conflicting descriptions in the methodology, including some missing fields or outdated names.
- **Formats and data types:** most fields are stored as TEXT, which is appropriate for the goal of retaining accurate information from source. However, this hinders the possibility of carrying out data validation on the database. A solution that would achieve both goals is to store the processed data in the database, as a separate table with its own schema.
- **Data processing:** the data processing script provided as a Jupyter Notebook works correctly but improvements could be made to the readability, documentation, and general formatting. Similar observations apply to the graphs it generates, which may benefit from an amended colour palette, the addition of gridlines, and a lower information density, depending on the ultimate purpose.

Aside from the points for improvement, the data structure represents strong improvement:

- The database has NOT NULL constraints.
- Foreign keys are implemented and enforced.
- Documentation is holistic and clear (aside from noted items).
- Data types are appropriate.

6.6 Source data

A sample check was carried out to compare the data in the register with the original sources, identified by the URLs provided in the database. These checks produced the following observations:

- It is not fully clarified how data is treated in cases where figures are reported in vague terms, e.g., “more than X Mt”.
- There are some minor inconsistencies between methodology document and the register on how financial and solar years are treated.
- Some incorrect or outdated entries were found. This may be the result of the original sources being updated by the publishers, which further enhances the points made in Section 6.3.4 regarding data source archiving.

7 FINDINGS AND RECOMMENDATIONS

7.1 Findings

The register has been evaluated against the data quality principles established in the previous review, and summarised in Section 5.2. The main findings against each principle are summarised in the following points:

- **Completeness** is strong and has improved significantly through the introduction of a structured database, source prioritisation fields, and a three-layered process for identifying CO₂ storage projects globally. The methodology document acknowledges that complete coverage and completeness of information cannot be guaranteed due to limitations in publicly available information. Data validation checks identified a few projects with less than two years of reported data and some missing source URLs, aspects that would benefit from ongoing monitoring. Therefore, carrying out the validation only during processing and not storing the processed data in the database could lead to inaccurate reporting, especially as the data volume increases in the future.
- **Understandability** is supported by a comprehensive methodology document, clear database structure, and documented processing workflow. Some limitations remain, including inconsistencies between the methodology and released materials, outdated references, incomplete data dictionaries and schema documentation, and opportunities to improve the readability of both the processing code and data visualisations.
- **Traceability** is high thanks to source values being retained in their original form, release logs being included in the database, source URLs being provided, and data processing being reproducible through the published script. Areas for improvement include the absence of detailed version-to-version change records, inconsistencies in documentation, and the fact that the source archive is not yet publicly accessible.
- **Accuracy** benefits from the separation of source and processed data, source prioritisation rules, four-eyes review procedures, and automated validation checks. Sample testing identified a small number of incorrect or outdated entries, some ambiguity in the treatment of imprecisely reported values, and minor inconsistencies in documented assumptions. The current validation framework provides a solid foundation but could be expanded and applied directly to the database to strengthen quality assurance throughout the register lifecycle. Furthermore, the implication of confidence in the data sources, including the widely different reporting standards remains unresolved, and perhaps worthy of being addressed separately.

7.2 Archiving data

The source data used to compile the register is currently only available to the public from the original publisher, if at all. It is therefore subject to change and/or removal, at the publisher's discretion. To support transparency, auditability, and future automation, it is recommended that the source files are stored in a public repository and indexed in a way that maintains a robust link to the data in the register.

To achieve this, DNV recommends implementing a two-tier storage approach: the original source documents should be stored in a public document repository as immutable records separated from the database, while being referenced by the register data within the database. Each record in the register should include a reference to the unique ID of the source document from which it was derived, analogous to the current "url" field, enabling users to trace information back to its originating evidence. The document metadata (repository ID, publication date and retrieval date, and source organisation) can be stored within the existing "source" table, ensuring that the provenance of all information is preserved, see Table 7-1. Public releases of the register can then be distributed alongside the corresponding source documents and supporting metadata, providing a transparent and auditable record of the information used in its compilation.

Where data sources are sensitive and cannot be published, this process can still be followed, but with a redaction of the source document and the ID maintained.

Table 7-1: Recommended new schema for "source" table. New fields are highlighted in blue.

Field	Type
source_id [PK]	TEXT
source_name	VARCHAR(200)
source_type	VARCHAR(100)
url	VARCHAR(500)
archived_name	VARCHAR(200)
source_notes	TEXT
document_id	TEXT
source_org	TEXT
retrieval_date	DATE
publication_date	DATE

7.3 Organisational Maturity

The improvements made to the register over the past cycle provide strong evidence of alignment against the core data quality principles. As detailed in Section 6.2, the previous recommendations have been addressed, and the enhancement of the register provides sufficient evidence of an increased organisational capability level, as defined in DNV-RP-0497 and DNV-RP-A204 /16/.

In the previous cycle, the project was at Level 1: "Initial", with limited formal governance allowing quality monitoring and measurement. Based on the evidence reviewed, the register now demonstrates organisational maturity characteristics consistent with Level 2: "Repeatable", with some aspects exhibiting characteristics of Level 3: "Defined". The organisational maturity framework is provided in Appendix C – Organisation Maturity.

Further progression towards Level 3: "Defined" and Level 4: "Managed" maturity will require enhanced validation controls, improved quality assurance, and source-data traceability. The recommendations in Table 7-2 are intended to support this progression.

These recommendations reflect the observations noted in section 6, with some supplementary information in section 7.2.

Table 7-2: List of recommendations.

ID	Recommendation	Priority
1	Processed data should be saved into a new table(s) within the database, which will also facilitate Recommendation 2.	Should
2	Introduce data validation rules in the database, running whenever edited, applying the quantitative rules provided in Appendix A, as appropriate.	Should

3	Enhance the documentation of the database extraction process (from SQL Server to SQLite, with any data type changes that may happen consequently), and the csv extraction process.	Must
4	Ensure cases where reported values are not straightforward annual rates (e.g. “more than X Mt”) are documented in the methodology, or source observations. This also applies to declared values referring to gross CO ₂ injection or net storage, and to how these impact the results.	Must
5	Apply coding best practice to data processing script: improve readability with comments and docstrings based on common standard (Google, Numpy, etc.), split into functions to avoid code duplication, log outputs into files, adopt a version-control tool, avoid in-function imports, and document all steps more thoroughly in the methodology document.	Should
6	Add lookup tables for fields that require explanation, like period_type, so database documentation is self-contained.	Could
7	Add a source document archive with indices linking documents to the data for repeatability and traceability.	Should
8	Actively use the results of data validation to guide a further round of Quality Assurance., i.e., focus checks on areas flagged during data validation.	Could
9	Seek evaluation of the relevant reporting standards as relevant to storage volume definition and incorporate this into confidence rating for the register data.	Should
10	Evaluate presentation format of graphs used in the annual report to maximise readability. For example, using stacked bar charts for composition, adding gridlines, softer colour palettes, reducing density of information in a single graph.	Could
11	Adopt dynamic cross-referencing in all text documents to prevent internal references from falling out of date.	Should
12	Ensure that all roles within the project are defined and consistent across all documents.	Must
13	In the Methodology document, provide a complete data dictionary for the latest database structure and a definitive list of public artefacts.	Should
14	Explain how data gaps for yet-unpublished data are filled, as well as the associated assumptions/risks.	Must

8 REFERENCES

- /1/ DNV, Data assurance summary, Doc No. 3093337, Report number 10591504-01, Rev. 0, 12/11/2025
- /2/ DNV RP 0497, Assurance of data quality management. October 2023
- /3/ Gao, X., & Krevor, S. (2025). The London Register of Subsurface CO₂ Storage: 2025 Annual Report. Imperial College London.
- /4/ ISO 8000. Data quality
- /5/ DNV (2025). Energy Transition Outlook “CCS to 2050”
- /6/ “3-London_Register_Methodology_v6.docx” provided by Xiaowei Gao, 2026-07-16
- /7/ “1-co2register_v5_0713.db” provided by Xiaowei Gao, 2026-07-16
- /8/ “2-AnnualReport2026-v6.docx” provided by Xiaowei Gao, 2026-07-16
- /9/ “4_London_Register_CO2_stored_by_project_year_v5_2026.xlsx” provided by Xiaowei Gao, 2026-07-16
- /10/ “4-Response-to-DNV-v1-0708.docx” provided by Xiaowei Gao, 2026-07-16
- /11/ “log_v5_0713.csv” provided by Xiaowei Gao, 2026-07-16
- /12/ “project_v5_0713.csv” provided by Xiaowei Gao, 2026-07-16
- /13/ “raw_observation_v5_0713.csv” provided by Xiaowei Gao, 2026-07-16
- /14/ “source_v5_0713.csv” provided by Xiaowei Gao, 2026-07-16
- /15/ “5_REPRODUCE_FROM_REGISTER_v2_0713.ipynb” provided by Xiaowei Gao, 2026-07-16
- /16/ DNV-RP-A204. Assurance of Digital Twins. October 2023
- /17/ ISO 25012:2008 Software engineering — Software product Quality Requirements and Evaluation (SQuaRE) — Data quality model

APPENDIX A – DATA VALIDATION RULES

ID	Rule	Description	Requirement
R-01	No future years: Data should not exist for years beyond current year	$\text{Max}(\text{Year}) \leq \text{Current Year}$	Accuracy
R-02	No non-zero values before Start_Year	Values < Start_Year = 0 or NaN	Accuracy
R-03	All values must be zero or positive	Value ≥ 0	Accuracy
R-04	Realistic change in CO ₂ storage value year to year	$\Delta \text{ year-on-year} \leq \text{plausible bound (e.g., <10 Mt or <500\%)}$	Accuracy
R-05	Change in CO ₂ storage value shouldn't be negative from year to year	Annual injection ≥ 0 ; cumulative monotonic	Accuracy
R-06	Primary keys are unique within each table	0 duplicates on primary key	Completeness
R-07	Fields "raw_id", "log_id", "project_id", "source_id", "period_type", "start_year", "end_year", "raw_value", "raw_unit", "priority_tier", "standards_text", and "storage_type" are not null or empty	100% completeness	Completeness
R-08	No duplicate names: each "project_name" should be unique within the "project" table. The same applies to "source_name" within the "source" table.	Project names normalised (case/whitespace) = unique	Completeness
R-09	No rows with only NaN values: There should be no rows where all columns are NaN	0 rows with all NaN across data fields	Completeness
R-10	Flag if there is a gap in data/CO ₂ storage values recorded	Missing year(s) within active range flagged	Completeness
R-11	Each "project_id" should have non-null "raw_value" for a minimum number of years (e.g. at least 3 years) to be considered valid	≥ 3 years with non-null values	Completeness
R-12	Years columns must be numeric: All year columns (1996-2024) should contain only numeric values or NaN	Year field = 4-digit integer; values numeric or NaN	Understandability
R-13	Valid list of values: values of certain fields should match a predefined list	Such fields are "raw_unit", "country", "source_type", "storage_type", "period_type", "priority_tier", "is_double_checked", and "is_storage_data_fixed"	Understandability

APPENDIX B – DATA QUALITY CHECKS

ID	Rule	Description	Requirement
C-01	Data quality validation	Apply data validation rules to database	Completeness
C-02	Review project search strategy	Evaluate process used to identify all active storage projects	Completeness
C-03	Review of improvements from previous cycle	Validation of claimed improvements	Traceability
C-05	Tracking report values to source	Replicate a sample of the calculations that produced the values in the report to see if they match	Traceability
C-06	Sample check of data sources	Check samples of data to see if they match the indicated source	Traceability
C-07	Expert review of methodology	Subject-matter expert review the register's methodology to flag any concerns	Traceability
C-08	Review structure of database	Review the database structure against good data engineering practice	Understandability
C-10	Compare shared data in the two formats provided	Data in .db file = data in .xlsx files	Understandability
C-11	Review data typing in database	Check whether all fields were assigned appropriate data types	Understandability
C-12	Evaluate database type selection	Review database technology use (SQLite, SQL Server, etc.)	Understandability
C-13	Verify adherence to methodology	Check whether the documented methodology is clear and evidence in the register products.	Understandability
C-14	Tables relate to each other via matching primary and foreign keys	Tables can be cross-referenced through unique keys	Understandability
C-15	Years are assigned as described in methodology document	When data is reported for a financial year, the reference year is the one featured in the source title.	Understandability

APPENDIX C – ORGANISATION MATURITY

Table C-1: Organisation Maturity.

AREAS LEVELS	 GOVERNANCE	 ORGANIZATION AND PEOPLE	 PROCESSES	 PROCESS EFFICIENCY	 REQUIREMENT DEFINITION	 METRICS AND DIMENSIONS	 ARCHITECTURE, TOOLS AND TECHNOLOGIES	 DATA STANDARDS
LEVEL 5: Optimized	Data quality management policies governs and drives data quality activities across the organization	Data quality management board oversees data quality activities across the organization	Improvement and response activities performed continuously and transparently	Immediate response to data quality issues with critical business impact	Data quality requirements used to drive measurement of business performance	Framework for metrics and dimensions are implemented across organization and metrics are connected to business KPIs	Data quality tools support self-service, change management and process improvements	Standards are implemented at the organizational level and compliance is monitored
LEVEL 4: Managed	Data quality management system defines policies in relation to business objectives	Skillset extended to include risk analysis of quality issues aligned with business objectives	Impact analysis, risk management and continuous improvement in place	Data quality metrics are linked to business criticality and continuous improvement	Data quality management system specifies requirements linked to improvement process	Metrics are linked to business impact, risk analysis and continuous improvement	Tools are driven by business objectives and include support for root cause analysis and risk mgmt.	Standards are used actively to reduce risk for critical business operations
LEVEL 3: Defined	Policies, roles and responsibilities defined at project level	Mandates, communications lines and skills defined at project level	Data quality issues trigger processes defined at project level	Data quality metrics are monitored in advance of business impact	Requirements defined and communicated at project level	Framework for metrics and dimensions defined at project level	Architecture and tools supports data quality management at project level	Common standards, domain models and semantics used at project level
LEVEL 2: Repeatable	Local initiatives address the requirement for policies	Locally defined roles and some basic skills	Best practices in place but not used consistently	Generic metrics are monitored at point of impact	Local initiatives define requirements	Metrics are reused locally in projects	Tools and technologies used consistently in selected projects	Industry standards and domain models used selectively across projects
LEVEL 1: Initial	Only ad-hoc or temporal policies in place	No formally defined roles or skillset	Ad-hoc or reactive responses to quality issues	No baseline and no monitoring of quality issues	Re-engineering used to derive requirements	Project specific metrics	Tools are used ad-hoc per project	Ad-hoc and inconsistent use of standards





About DNV

DNV is the independent expert in risk management and assurance, operating in more than 100 countries. Through its broad experience and deep expertise DNV advances safety and sustainable performance, sets industry benchmarks, and inspires and invents solutions.

Whether assessing a new ship design, optimizing the performance of a wind farm, analyzing sensor data from a gas pipeline or certifying a food company's supply chain, DNV enables its customers and their stakeholders to make critical decisions with confidence.

Driven by its purpose, to safeguard life, property, and the environment, DNV helps tackle the challenges and global transformations facing its customers and the world today and is a trusted voice for many of the world's most successful and forward-thinking companies.